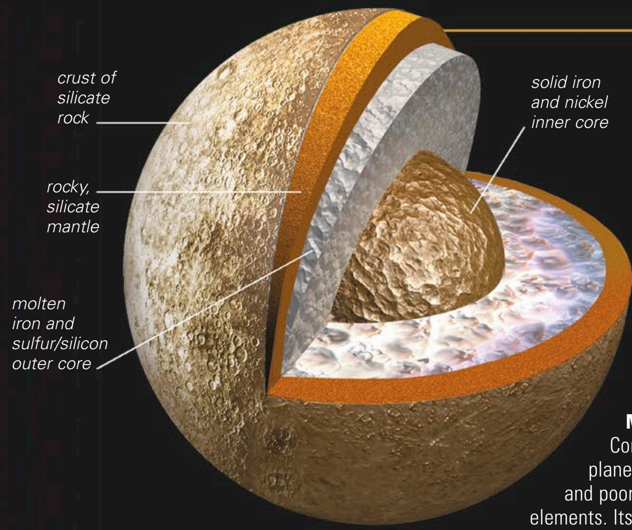


MERCURY INTERIOR
Compared to the other rocky planets, Mercury is very rich in metals and poor in heat-producing radioactive elements. Its huge iron core is probably solid.

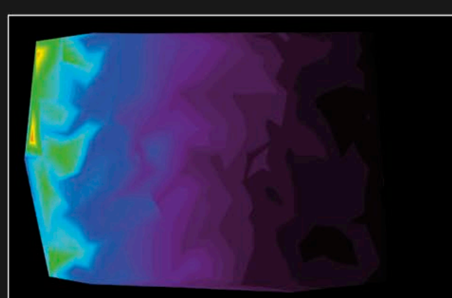
STRUCTURE

The very high density of Mercury indicates that it is rich in iron. This iron sank to the center some 4 billion years ago, producing a huge core 2,235 miles (3,600 km) in diameter. There is a possibility that a thin layer of the outer core is still molten. The solid rocky mantle is about 340 miles (550 km) thick and makes up most of the outer 25 percent of the planet. This outer mantle has slowly cooled, and during the last billion years, volcanic eruptions and lava flows have ceased, making the planet tectonically inactive. The mantle and the thin crust mainly consist of the silicate mineral anorthosite, just like the old lunar highlands. There are no iron oxides. Unlike on other planets, it seems that all the iron has gone into the core, which produces a magnetic field with a strength that is about 1 percent of Earth's magnetic field.

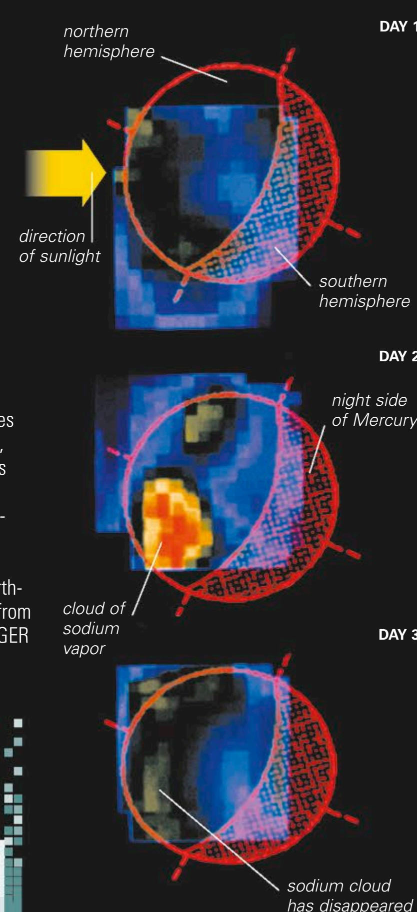
ATMOSPHERE

Mercury has a very thin temporary atmosphere because the planet's mass is too small for an atmosphere to persist. Mercury is very close to the Sun, so daytime temperatures are extremely high, reaching 800°F (430°C). The escape velocity is less than half that of Earth, so hot, light elements in the atmosphere, such as helium, quickly fly off into space. All the atmospheric gases therefore need constant replenishment. Mercury's atmosphere was analyzed by an ultraviolet spectrometer onboard the Mariner 10 spacecraft in 1974. Oxygen, helium, and hydrogen were detected in this way, and subsequently atmospheric sodium, potassium, and calcium have been detected by Earth-based telescopes. The hydrogen and helium are captured from the solar wind of gas that is constantly escaping from the Sun. The other elements originate from the planet's surface and are intermittently kicked up into the tenuous atmosphere by the impact of ions from Mercury's magnetosphere and micrometeorite particles from the solar system dust cloud. The atmospheric gases are much denser on the cold night side of the planet than on the hot day side, as the molecules have less energy to escape.

ATMOSPHERIC COMPOSITION
Oxygen is the most abundant gas, followed by sodium and helium. However, loss and regeneration of the gases is continuous, and the atmospheric composition can vary drastically over time.



MERCURY'S SODIUM TAIL
Pressure exerted by sunlight pushes sodium atoms away from Mercury, forming a "tail" some 25,000 miles (40,000 km) long. Mercury and the Sun are off to the left in this false-colored view of Mercury's sodium tail. Emissions from this tail have previously been observed with Earth-based telescopes, but this image from a spectrometer onboard MESSENGER is the most detailed image yet.



TEMPORARY ATMOSPHERE
Thin clouds of sodium suddenly appear over some regions of Mercury and then just as quickly disappear, as seen in these false-color observations made by the Kitt Peak Solar Observatory in the US. The clouds might be produced by meteorite impacts—the freshly cratered surface releases sodium vapor when it is next heated by sunlight. Another possibility is that ionized particles actually hit Mercury's surface and release sodium from the regolith.

MERCURY PROFILE

AVERAGE DISTANCE FROM THE SUN 36 million miles (57.9 million km)		**ROTATION PERIOD** 59 Earth days	
SURFACE TEMPERATURE –290°F to 810°F (–180°C to 430°C)		**ORBITAL PERIOD (LENGTH OF YEAR)** 88 Earth days	
DIAMETER	3,029 miles (4,875 km)	**MASS (EARTH = 1)**	0.055
VOLUME (EARTH = 1)	0.056	**GRAVITY AT EQUATOR (EARTH = 1)**	0.38
NUMBER OF MOONS	0	**SIZE COMPARISON**	

OBSERVATION
Never more than 28° away from the Sun in the sky, Mercury is always seen at dawn or dusk. It is the most difficult of the nearby planets to spot and is visible only for a few days each month.